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Fruit Classification Convolutional Neural Network

I am Trevor Streng, and I will be working on this project solo. I am planning on using the Fruit Classification dataset from Kaggle.com.

(<https://www.kaggle.com/datasets/icebearogo/fruit-classification-dataset>) This dataset includes 101 classes of different fruits with 50,000 images. It is split into train, test, and validation sets. Each folder is split into subfolders of corresponding classes. The dataset seems to be clean, but we will need to resize the images so they are consistent.

I am planning to build a Convolutional Neural Network to classify the different types of fruits. I will start with the topics we have learned in class, but will research additional advanced topics to maximize my results while still being able to run the model on my available hardware.

To reduce the risk of overfitting, I will use batch normalization as well as augmentation techniques, like image rotation.

My goal for this project is to achieve high test and validation accuracy and learn more about different patterns that produce good results in image classification.

My expected outcome for this project is to try to classify images with 85% or greater accuracy while reducing training times and hardware constraints. I would also like to learn more about what produces better results without using too many parameters. For example, keeping the parameters a bit under 20,000,000.

My initial planning will consist of first, researching, then, designing and prototyping a model, third, testing, analyzing, and refactoring the model, fourth, creating useful visualizations, and finally, creating a PowerPoint presentation and video recording.

For a timeline I am thinking, research should be done by July 7th. Second, designing and prototyping the model completed by July 11th. Third, testing, analyzing, and refactoring to be finished by July 16th. Fourth, creating useful visualizations by July 19th. Finally, creating the PowerPoint presentation and recording finished by July 24th.

I will start with researching topics and adding them to a document. Next, I will create a design and implement it into a prototype. Once the prototype is completed, I could use an iterative approach where I will test, analyze, and refactor to improve the model. Fourth, I will create different visualizations. For example, a chart of which classes the model struggled with vs which on it performed well on. Another chart for validation vs training accuracy. A visualization of the model's architecture, some images

that were classified correctly and incorrectly and an image augmentation example. Finally, I will create a PowerPoint presentation and record myself describing each slide. For example, I will go over the model's architecture, the parameters and layers, how different layers are implemented, and describe how augmentations were implemented. I will also show the results, how I analyzed and improved the model through the iterative process, and specify changes I made to the model and how they affected the results. And finally, I will go over the environment and libraries I used and how to run the model.